

2. Social Engineering Attack Analysis

Introduction

Social engineering is a cyberattack technique that manipulates people into revealing information or performing actions that compromise security. Instead of attacking systems directly, attackers target human psychology. They rely on emotions such as fear, curiosity, trust, urgency, or greed to influence the victim's decisions.

This topic is important because many cyber incidents begin with a simple social engineering trick. Once the victim is convinced, the attacker may gain access to accounts, devices, or private data.

Objective

The objective of this exercise is to understand what social engineering is, how it works, what types of attacks exist, and how users can protect themselves.

What is Social Engineering?

Social engineering is the act of deceiving a person into revealing sensitive information or taking an unsafe action. Attackers often pretend to be trusted people such as bank officers, technical support staff, government agents, or company employees.

The attack succeeds when the victim believes the attacker and responds without proper verification.

Common Types of Social Engineering Attacks

1. Phishing

Phishing is a fake email or website attack used to steal credentials, banking data, or other sensitive information.

2. Smishing

Smishing is phishing through SMS. The attacker sends a text message that contains a malicious link or fake request.

3. Vishing

Vishing is voice phishing through phone calls. The attacker uses a phone call to impersonate a trusted organization.

4. QRishing

QRishing uses malicious QR codes to redirect victims to fake websites or malware downloads.

5. Pretexting

Pretexting happens when the attacker creates a fake identity or scenario to gain access to information.

6. Tailgating

Tailgating is a physical attack where an unauthorized person follows someone into a restricted area.

7. Deepfake-based social engineering

Deepfake attacks use fake audio or video to impersonate a real person. This makes the scam more believable because the victim sees or hears someone they trust.

Why Social Engineering Works

Social engineering works because it attacks human behavior instead of software. People often respond quickly when they feel:

- afraid
- rushed
- curious
- pressured
- confident that the source is real

Attackers take advantage of these feelings to push the victim into making a mistake.

Example Scenario

A common example is a message that claims the victim's account has been suspended and must be verified immediately. The message may include a fake login link. If the victim enters their details, the attacker steals the credentials.

Another example is a fake support call asking for OTP verification. If the victim gives the OTP, the attacker may take over the account.

Impact of Social Engineering

Social engineering attacks can cause:

- stolen passwords
- financial loss
- identity theft
- account takeover
- malware infection
- data leakage
- damage to reputation

For organizations, the impact can also include business disruption and loss of customer trust.

Prevention

To reduce the risk of social engineering:

- verify identity before sharing information
- do not click suspicious links
- do not trust urgent messages immediately
- check email headers and URLs carefully
- avoid scanning unknown QR codes
- use MFA
- report suspicious messages

- follow security awareness training

Conclusion

Social engineering is one of the most effective cyberattack methods because it exploits human weakness. It includes many forms such as phishing, smishing, vishing, QRishing, pretexting, and deepfake scams. The best protection is awareness, verification, and a habit of not trusting messages blindly.